In [111]:
```
# PROBLEM STATEMENT : Heart Disease Dataset
```

In [ ]:
```

```

In [ ]:
```
# 1. Load, Explore and Clean the Dataset
```

In [4]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv(r"C:\Users\Administrator\Downloads\Heart_Disease_UCI.csv")
df
```

Out[4]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	tl
0	69	1	0	160	234	1	2	131	0	0.1	1	1	
1	69	0	0	140	239	0	0	151	0	1.8	0	2	
2	66	0	0	150	226	0	0	114	0	2.6	2	0	
3	65	1	0	138	282	1	2	174	0	1.4	1	1	
4	64	1	0	110	211	0	2	144	1	1.8	1	0	
...	...	...	...	...	...	...	...	...	...	...	...	...	
292	40	1	3	152	223	0	0	181	0	0.0	0	0	
293	39	1	3	118	219	0	0	140	0	1.2	1	0	
294	35	1	3	120	198	0	0	130	1	1.6	1	0	
295	35	0	3	138	183	0	0	182	0	1.4	0	0	
296	35	1	3	126	282	0	2	156	1	0.0	0	0	

297 rows × 14 columns

In [5]:
```
df.head(10)
```

Out[5]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal
0	69	1	0	160	234	1	2	131	0	0.1	1	1	0
1	69	0	0	140	239	0	0	151	0	1.8	0	2	0
2	66	0	0	150	226	0	0	114	0	2.6	2	0	0
3	65	1	0	138	282	1	2	174	0	1.4	1	1	0
4	64	1	0	110	211	0	2	144	1	1.8	1	0	0
5	64	1	0	170	227	0	2	155	0	0.6	1	0	2
6	63	1	0	145	233	1	2	150	0	2.3	2	0	1
7	61	1	0	134	234	0	0	145	0	2.6	1	2	0
8	60	0	0	150	240	0	0	171	0	0.9	0	0	0
9	59	1	0	178	270	0	2	145	0	4.2	2	0	2

In [6]:
```
df.describe()
```

Out[6]:

	age	sex	cp	trestbps	chol	fbs	reste
count	297.000000	297.000000	297.000000	297.000000	297.000000	297.000000	297.0000
mean	54.542088	0.676768	2.158249	131.693603	247.350168	0.144781	0.9966
std	9.049736	0.468500	0.964859	17.762806	51.997583	0.352474	0.9949
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.0000
25%	48.000000	0.000000	2.000000	120.000000	211.000000	0.000000	0.0000
50%	56.000000	1.000000	2.000000	130.000000	243.000000	0.000000	1.0000
75%	61.000000	1.000000	3.000000	140.000000	276.000000	0.000000	2.0000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.0000

In [7]:
```
print(df.isnull().sum())
```

```
age          0
sex          0
cp           0
trestbps     0
chol         0
fbs          0
restecg      0
thalach      0
exang        0
oldpeak      0
slope        0
ca           0
thal         0
condition    0
dtype: int64
```

In [8]:
```
print(df.duplicated().sum())
```

```
0
```

In [9]:
```
print(df.dtypes)
```

```
age            int64
sex            int64
cp             int64
trestbps       int64
chol           int64
fbs            int64
restecg        int64
thalach        int64
exang          int64
oldpeak      float64
slope          int64
ca             int64
thal           int64
condition      int64
dtype: object
```

In [10]:
```
print(df.shape)
```

```
(297, 14)
```

In [11]:
```
print(df.info())
```

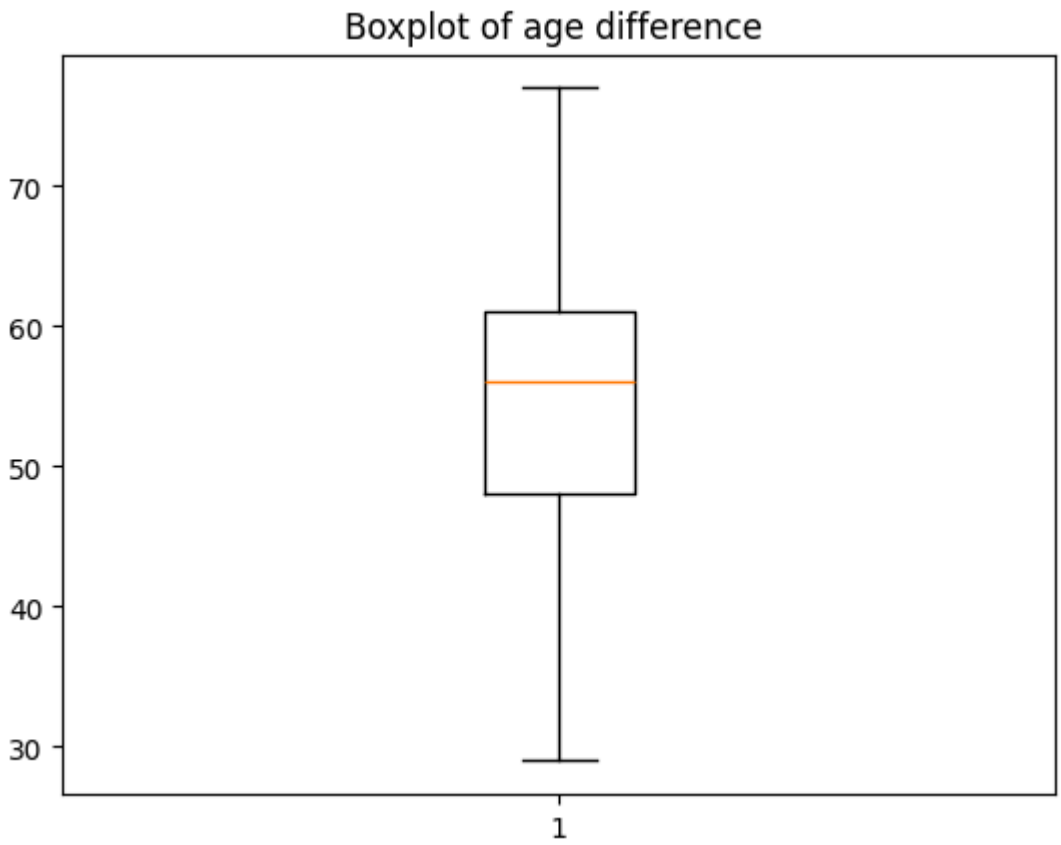
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 297 entries, 0 to 296
Data columns (total 14 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   age        297 non-null    int64
 1   sex        297 non-null    int64
 2   cp         297 non-null    int64
 3   trestbps   297 non-null    int64
 4   chol       297 non-null    int64
 5   fbs        297 non-null    int64
 6   restecg    297 non-null    int64
 7   thalach    297 non-null    int64
 8   exang      297 non-null    int64
 9   oldpeak    297 non-null    float64
 10  slope      297 non-null    int64
 11  ca         297 non-null    int64
 12  thal       297 non-null    int64
 13  condition  297 non-null    int64
dtypes: float64(1), int64(13)
memory usage: 32.6 KB
None
```

In [12]:
```
df.columns
```

Out[12]:
```
Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach',
       'exang', 'oldpeak', 'slope', 'ca', 'thal', 'condition'],
      dtype='object')
```

In [13]:
```
import matplotlib.pyplot as plt

plt.boxplot(df['age'])
```

```
plt.title("Boxplot of age difference ")
plt.show()
```



In [ ]:

In [ ]:

In [14]:
```
# 2. Gender Distribution Analysis
```

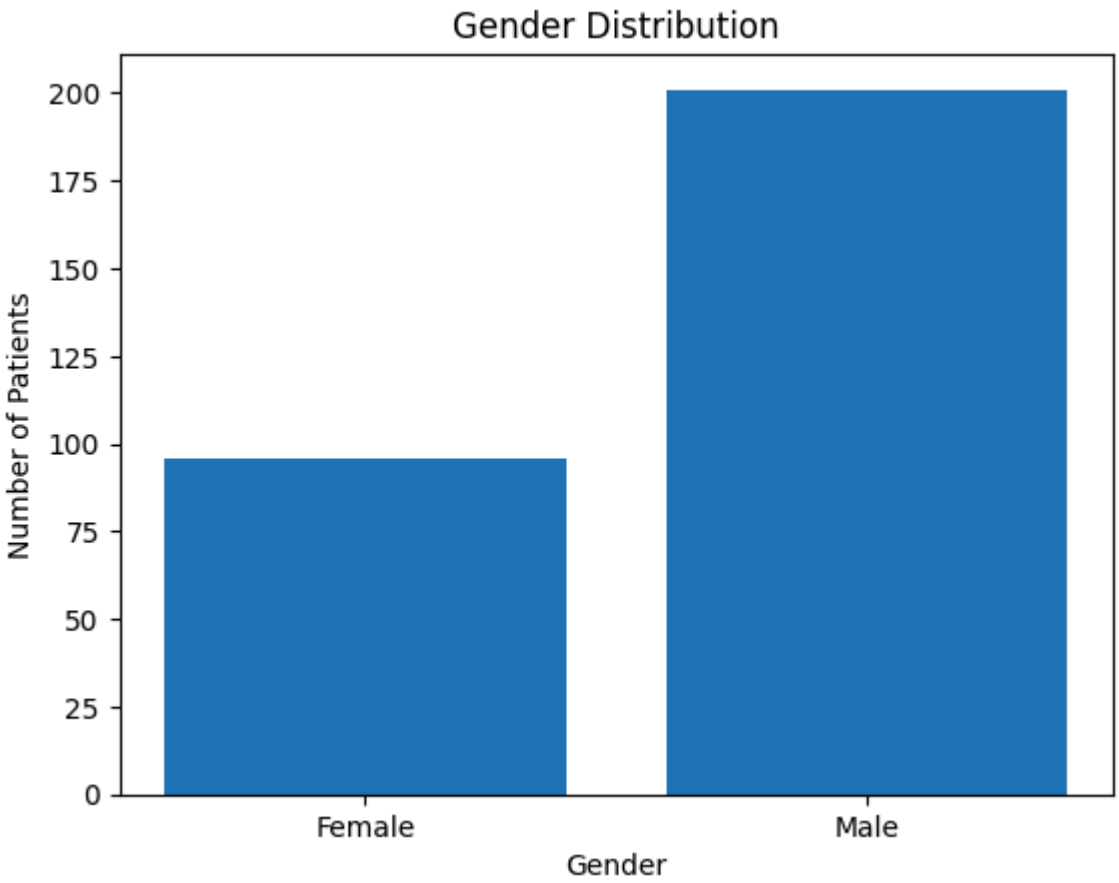
In [15]:
```
# Count number of males and females
total_males_female = df['sex'].sum()
print("Total number of males and females:", total_males_female)
```

```
Total number of males and females: 201
```

In [16]:
```
gender_counts = df['sex'].value_counts()

male = gender_counts[1]
female = gender_counts[0]

total = len(df)
male_percent = (male/total)*100
female_percent = (female/total)*100
print("Total number of males and females:", total_males_female)
print("Percentage of male and female present")
print("Male %:", male_percent)
print("Female %:", female_percent)

plt.figure()
plt.bar(['Female','Male'], [female, male])
plt.title("Gender Distribution")
plt.xlabel("Gender")
```

```
plt.ylabel("Number of Patients")
plt.show()
```

```
Total number of males and females: 201
Percentage of male and female present
Male %: 67.67676767676768
Female %: 32.323232323232325
```



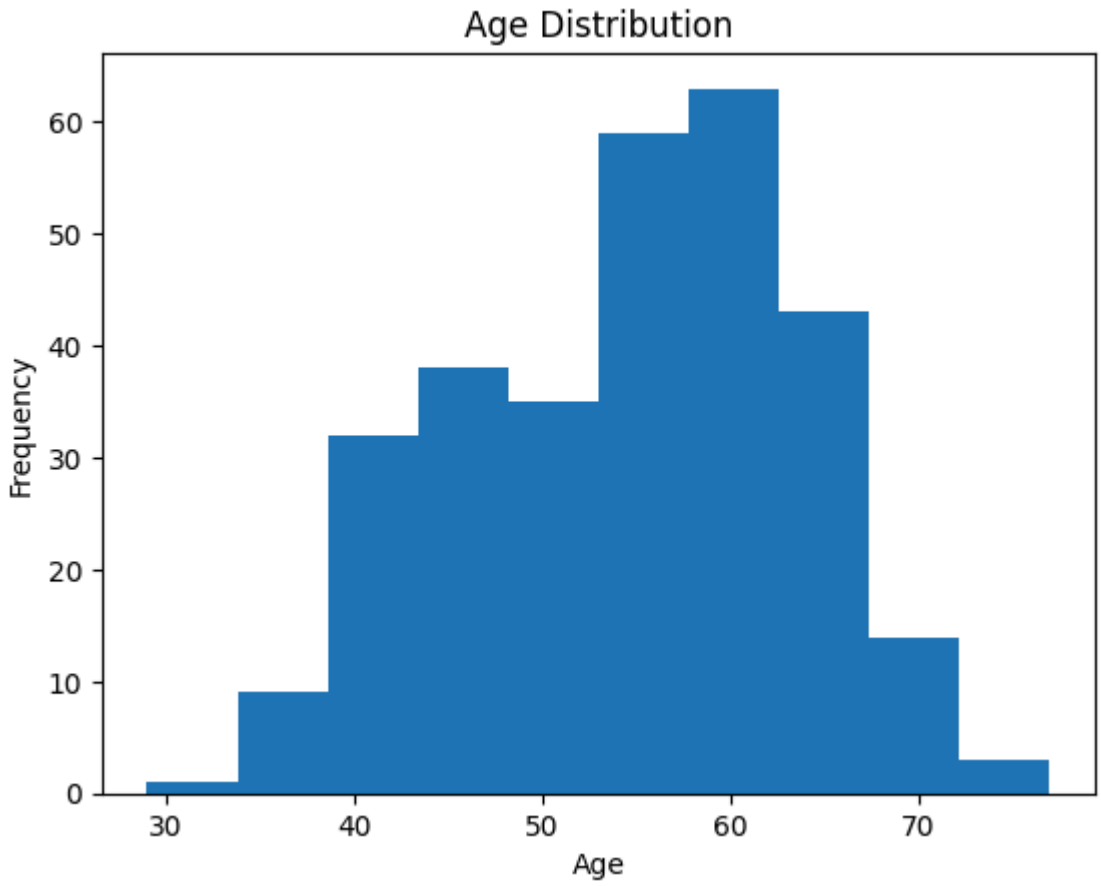
In [ ]:

In [ ]:

In [17]:
```
# Case 3 : Age Analysis:
```

In [18]:
```
print("check the age difference of all the people")
print("Minimum Age:", df['age'].min())
print("Maximum Age:", df['age'].max())
print("Mean Age:", df['age'].mean())
print("Median Age:", df['age'].median())

plt.figure()
plt.hist(df['age'], bins=10)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```

```
check the age difference of all the people
Minimum Age: 29
Maximum Age: 77
Mean Age: 54.54208754208754
Median Age: 56.0
```



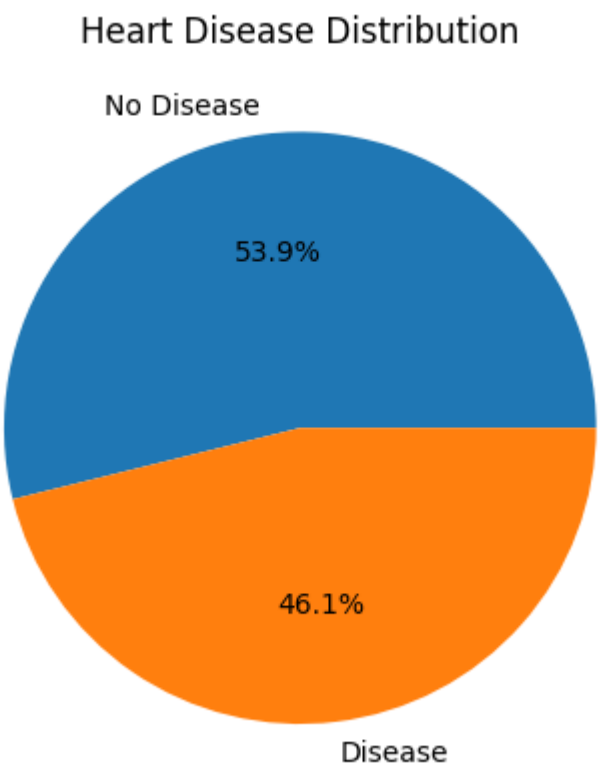
In [ ]:

In [ ]:

In [19]:
```
# case 4: Target variable analysis

# to check whearther the person is suffiring from disease or not
```

In [39]:
```
print("Check the person wellness")
target_counts = df['condition'].value_counts()
print(target_counts)
plt.figure()
plt.pie(target_counts, labels=["No Disease","Disease"], autopct="%1.1f%%")
plt.title("Heart Disease Distribution")
plt.show()
```

```
Check the person wellness
condition
0    160
1    137
Name: count, dtype: int64
```



In [ ]:

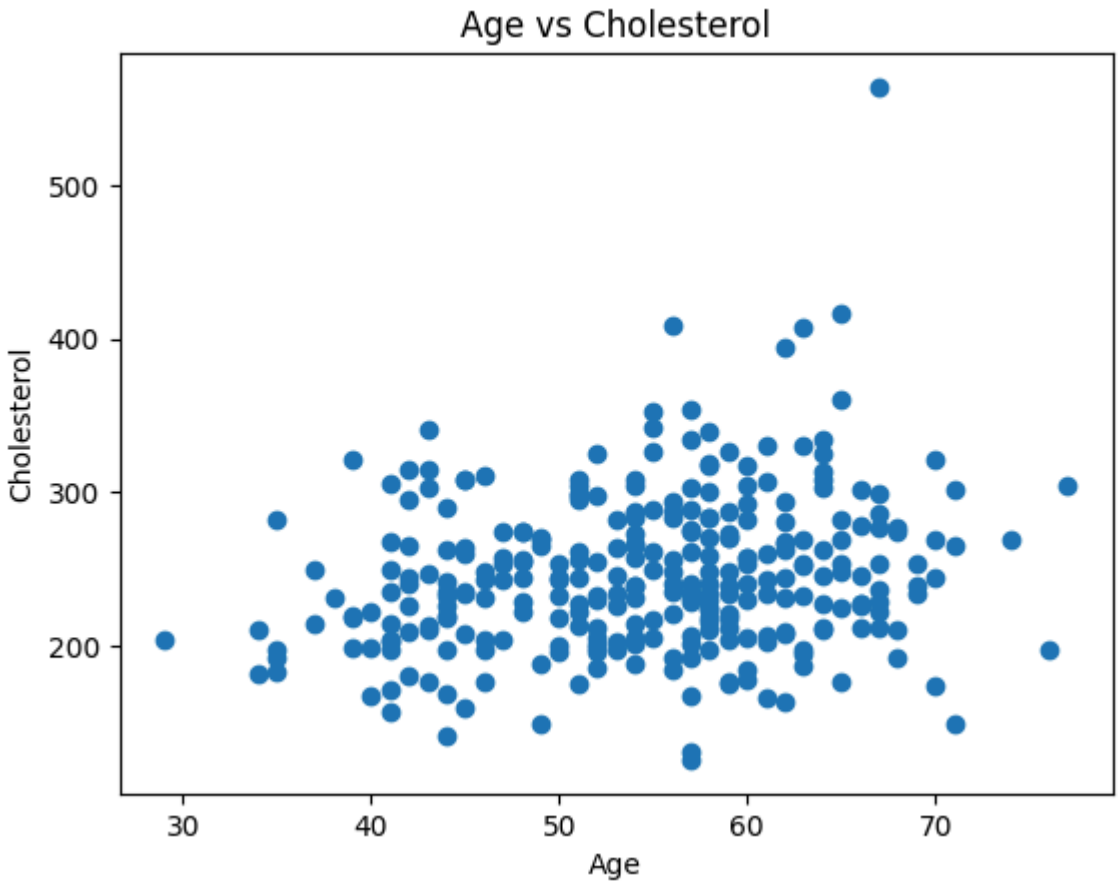
In [ ]:

In [21]:
```
# case 5: To find Correlation Between Age and Cholesterol
```

In [22]:
```
print("To check what age of person are suffring from cholestrol level ")
correlation = df[['age','chol']].corr()
print(correlation)
```

```
To check what age of person are suffring from cholestrol level
           age      chol
age   1.000000  0.202644
chol  0.202644  1.000000
```

In [23]:
```
plt.figure()
plt.scatter(df['age'], df['chol'])
plt.title("Age vs Cholesterol")
plt.xlabel("Age")
plt.ylabel("Cholesterol")
plt.show()

print("If correlation is zero means weak relatonship")
```




```
If correlation is zero means weak relatonship
```

In [44]:
```
# Interpret the relationship between Age and Cholesterol
age_chol_corr = df['age'].corr(df['chol'])
print(f"Correlation between Age and Cholesterol: {age_chol_corr:.2f}")
if abs(age_chol_corr) < 0.3:
    print("Interpretation: There is a weak relationship between age and cholesterc
elif abs(age_chol_corr) < 0.7:
    print("Interpretation: There is a moderate relationship between age and choles
else:
    print("Interpretation: There is a strong relationship between age and choleste
if age_chol_corr > 0:
    print("The relationship is positive, means as age increases, cholesterol tends
else:
    print("The relationship is negative, means as age increases, cholesterol tends
```

```
Correlation between Age and Cholesterol: 0.20
Interpretation: There is a weak relationship between age and cholesterol levels.
The relationship is positive, means as age increases, cholesterol tends to increas
e.
```

In [ ]:

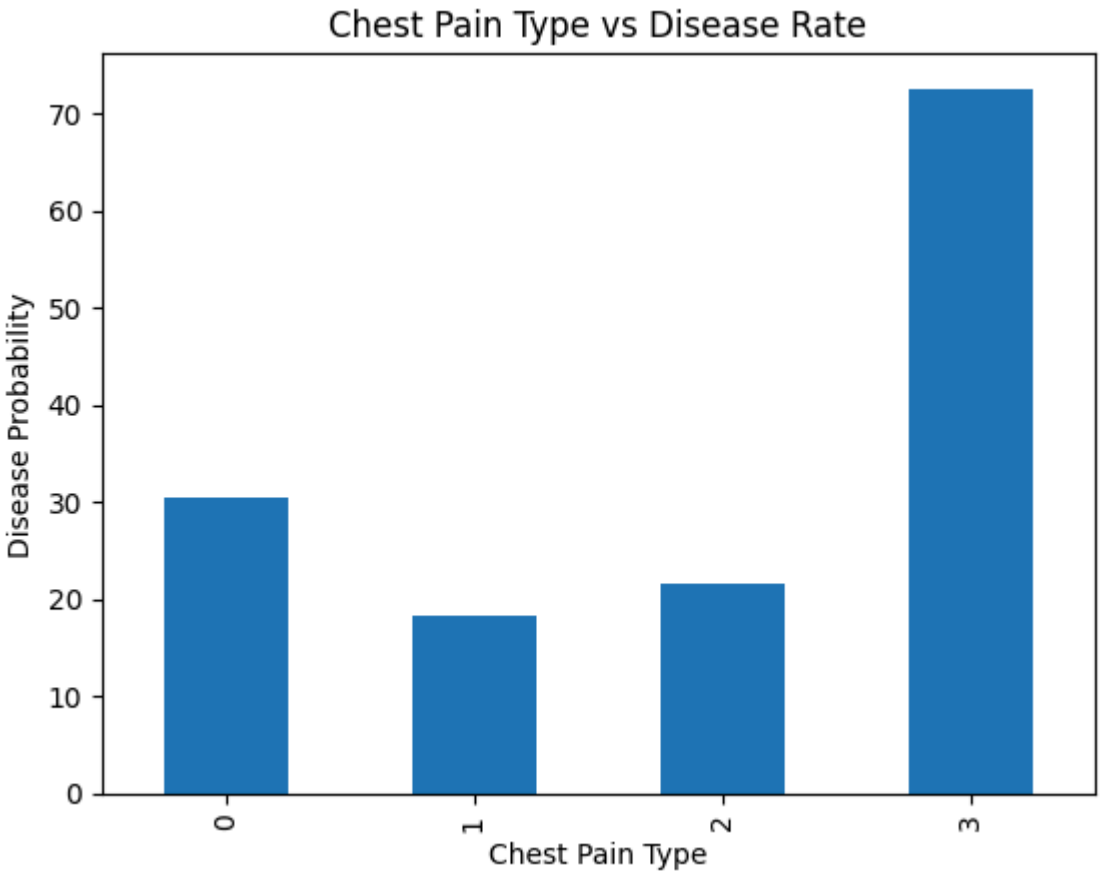
In [24]:
```
#case 6: TO check what probability of chest pain has the disease which can affect
```

In [40]:
```
print("to see the cp and there condition")
cp_group = df.groupby('cp')['condition'].mean()*100
print(cp_group)
```

```
to see the cp and there condition
cp
0    30.434783
1    18.367347
2    21.686747
3    72.535211
Name: condition, dtype: float64
```

In [41]:
```
print("TO check what probability of chest pain has the disease which can affect")
cp_group.plot(kind='bar')
plt.title("Chest Pain Type vs Disease Rate")
plt.xlabel("Chest Pain Type")
plt.ylabel("Disease Probability")
plt.show()
```

```
TO check what probability of chest pain has the disease which can affect
```



In [46]:
```
# Identify which chest pain type is most risky
most_risky_cp = cp_group.idxmax()
highest_rate = cp_group.max()
print(f"The chest pain type with the highest disease rate is {most_risky_cp} with
```

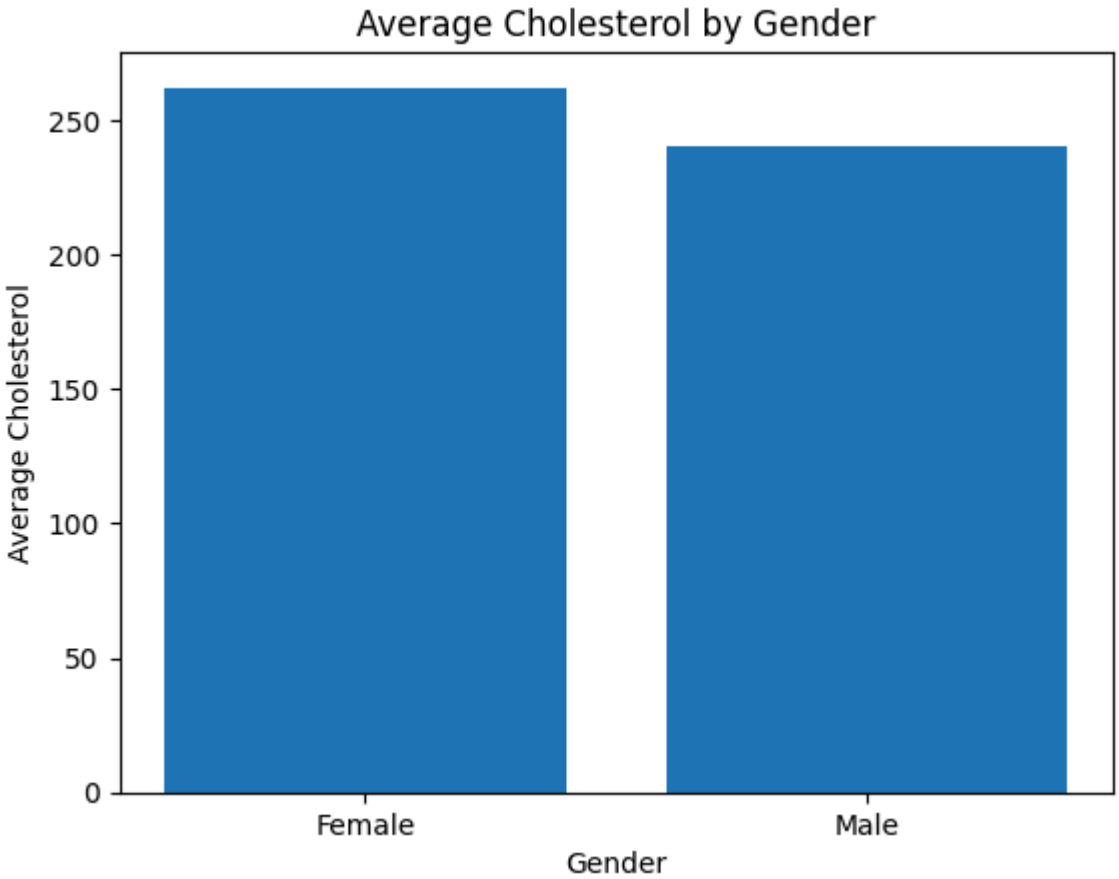
```
The chest pain type with the highest disease rate is 3 with a rate of 72.54%.
```

In [ ]:

In [42]:
```
#case 7: Average Cholesterol by Gender

chol_gender = df.groupby('sex')['chol'].mean()
print(chol_gender)
```

```
sex
0    262.229167
1    240.243781
Name: chol, dtype: float64
```

In [28]:
```
plt.figure()
plt.bar(['Female','Male'], chol_gender)
plt.title("Average Cholesterol by Gender")
plt.xlabel("Gender")

plt.ylabel("Average Cholesterol")
plt.show()
```



In [ ]:

In [ ]:

In [29]:
```
#case 8: Resting Blood Pressure Analysis
```

In [30]:
```
print("To check the average bp of the person")
avg_bp = df['trestbps'].mean()
print("Average BP:", avg_bp)
```

```
To check the average bp of the person
Average BP: 131.69360269360268
```

In [31]:
```
print("To check the persons with hign blood pressure")
high_bloodpressure = df[df['trestbps'] > 140]
print("High Blood Pressure Patients:", len(high_bloodpressure))
```

```
To check the persons with hign blood pressure
High Blood Pressure Patients: 66
```

In [38]:
```
print("Compare disease presence in high blood pressure group")
print("Disease % in High BP:",
      high_bloodpressure['condition'].mean()*100)
```

```
Compare disease presence in high blood pressure group
Disease % in High BP: 59.09090909090909
```

In [ ]:

In [ ]:

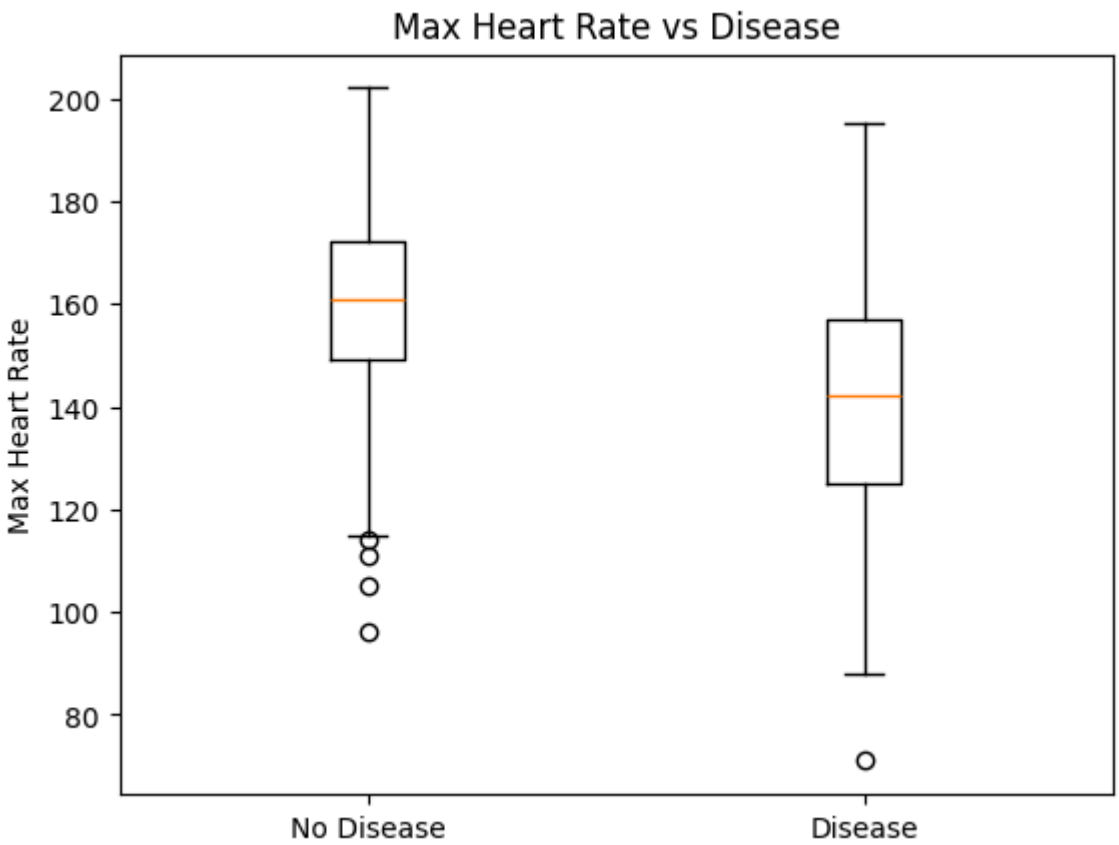
In [33]:
```
#Case 9 : To check the maximum heart rate vs Disease
```

In [62]:
```
print("Check the heart rate and accordingly the disease is present or not")
disease = df[df['condition']==1]['thalach']
no_disease = df[df['condition']==0]['thalach']

print("Avg HR Disease:", disease.mean())
print("Avg HR No Disease:", no_disease.mean())
```

```
Check the heart rate and accordingly the disease is present or not
Avg HR Disease: 139.1094890510949
Avg HR No Disease: 158.58125
```

In [61]:
```
print("TO check the expectional disease using outlier detection")
plt.figure()
plt.boxplot([no_disease, disease])
plt.xticks([1,2], ["No Disease","Disease"])
plt.title("Max Heart Rate vs Disease")
plt.ylabel("Max Heart Rate")
plt.show()
```

```
TO check the expectional disease using outlier detection
```



In [ ]:

In [ ]:

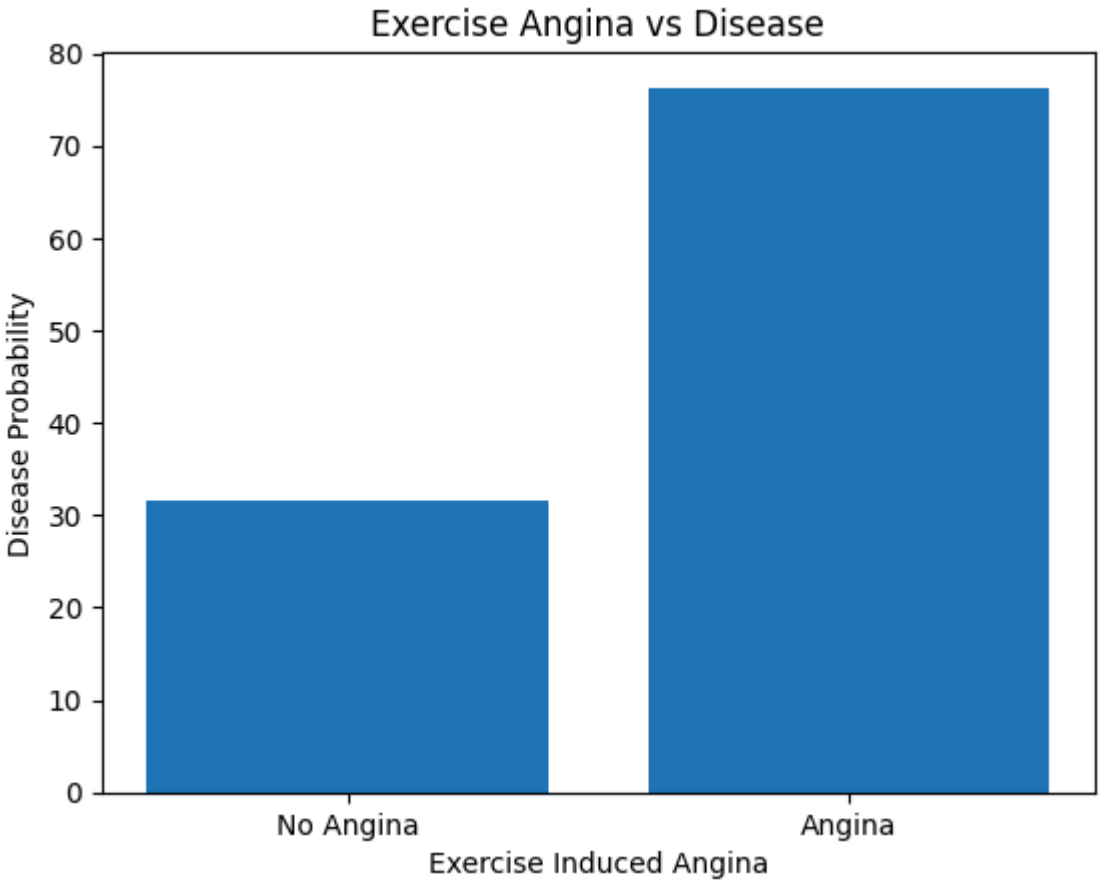
In [63]:
```
#Case 10 : Exercise Induced Angina Impact
```

In [43]:
```
exang_group = df.groupby('exang')['condition'].mean()*100
print(exang_group)
```

```
exang
0    31.50000
1    76.28866
Name: condition, dtype: float64
```

In [102]:
```
print("Impact of angina by doing exercise")
plt.figure()
plt.bar(['No Angina','Angina'], exang_group)
plt.title("Exercise Angina vs Disease")
plt.xlabel("Exercise Induced Angina")
plt.ylabel("Disease Probability")

plt.show()
```

```
Impact of angina by doing exercise
```



In [ ]:

In [ ]:

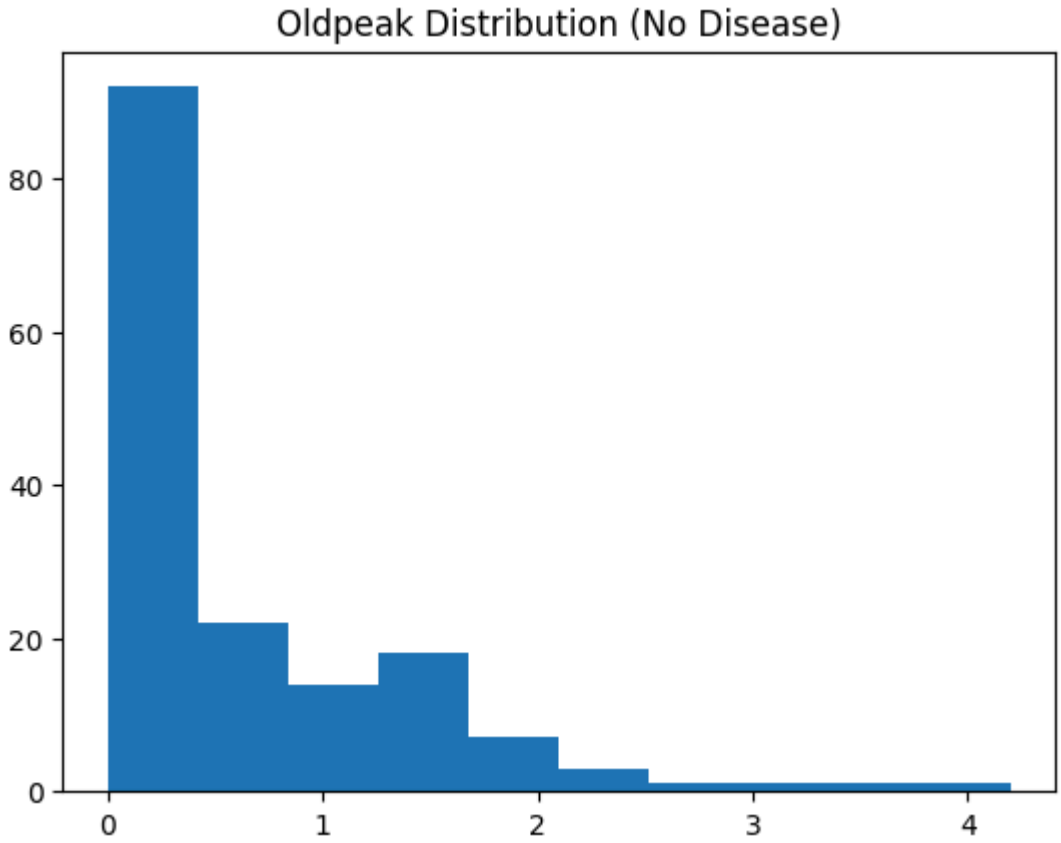
In [ ]:
```
#case 11 : ST Depression (oldpeak) Analysis
```

In [49]:
```
print("To see whether the person is suffering from ST depression or not")
oldpeak = df.groupby('condition')['oldpeak'].mean()
print(oldpeak)
```

```
To see whether the person is suffering from ST depression or not
condition
0    0.598750
1    1.589051
Name: oldpeak, dtype: float64
```

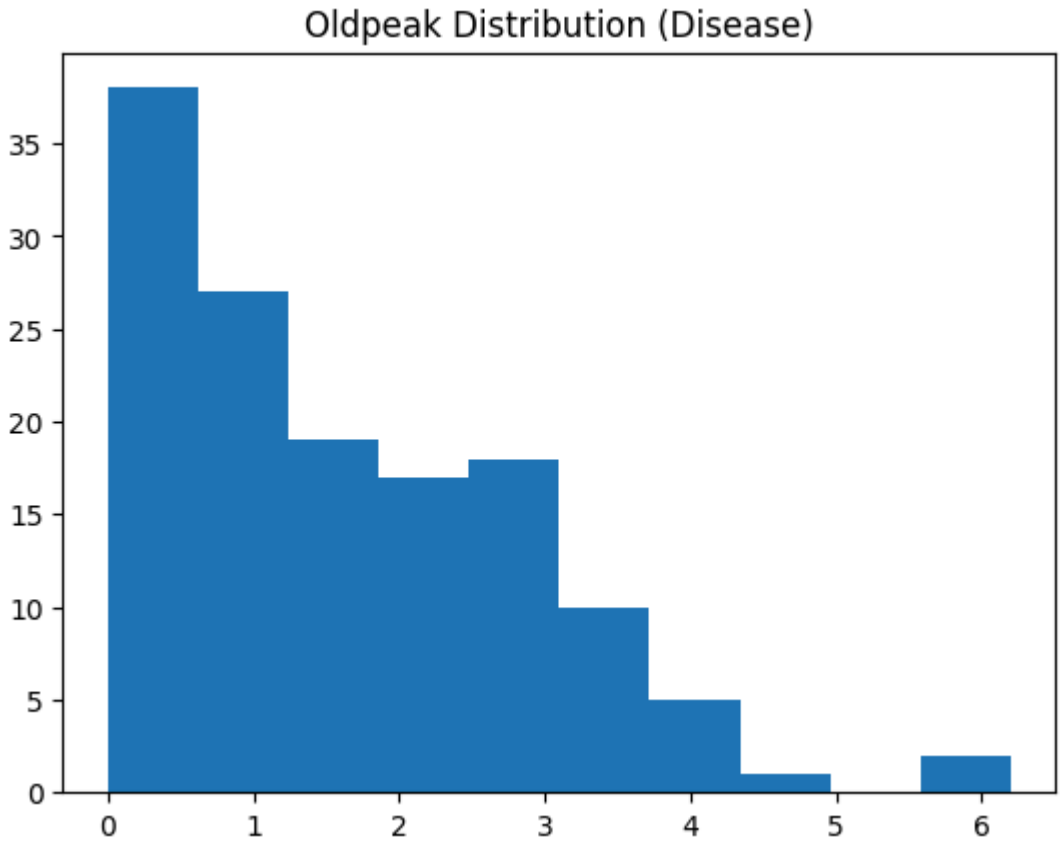
In [50]:
```
print("Persons having ST depression with no disease")
plt.figure()
plt.hist(df[df['condition']==0]['oldpeak'], bins=10)
plt.title("Oldpeak Distribution (No Disease)")
plt.show()
```

```
Persons having ST depression with no disease
```



In [51]:
```
print("Persons having ST Depression with disease")
plt.figure()
plt.hist(df[df['condition']==1]['oldpeak'], bins=10)
plt.title("Oldpeak Distribution (Disease)")
plt.show()
```

```
Persons having ST Depression with disease
```



In [52]:
```
# Identify trend
if oldpeak[1] > oldpeak[0]:
    print("Trend: Patients with heart disease tend to have higher ST depression (o
else:
    print("Trend: Patients with heart disease tend to have lower ST depression (ol
```

```
Trend: Patients with heart disease tend to have higher ST depression (oldpeak)
```

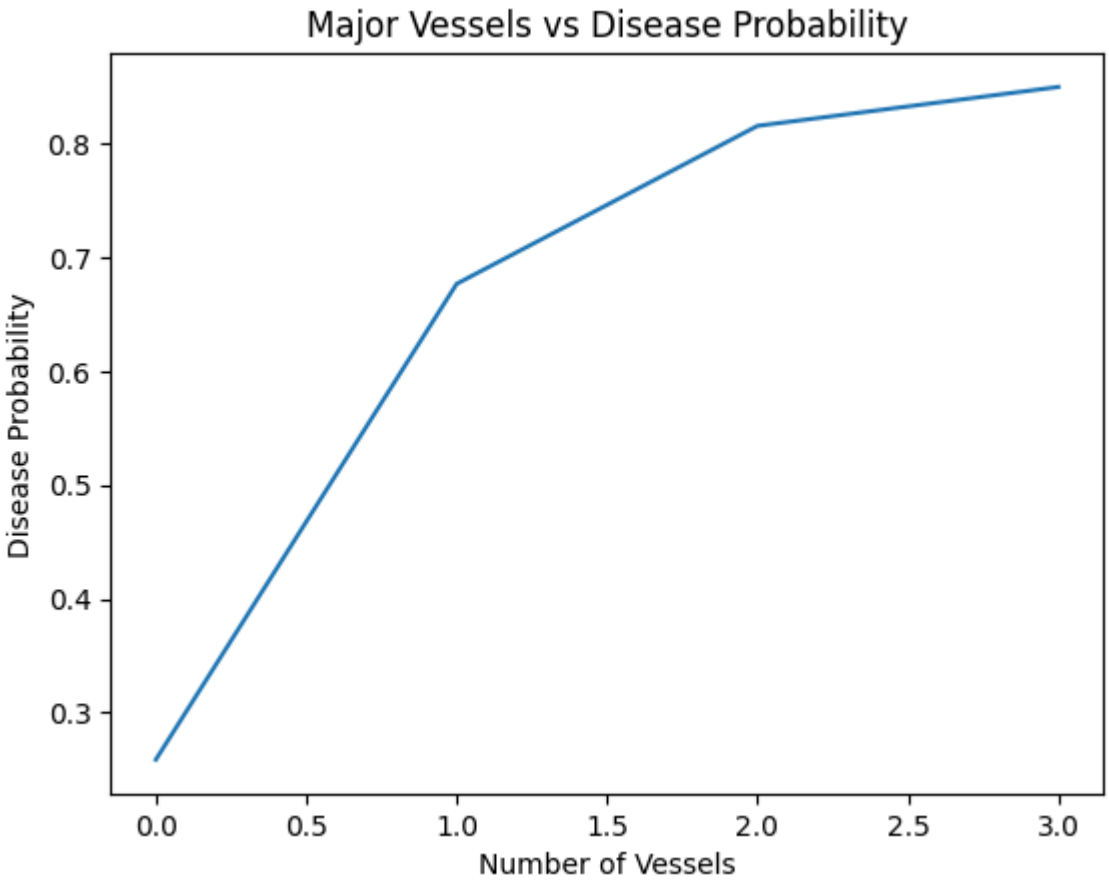
In [ ]:

In [ ]:
```
#CASE 12: Number of Major Vessels (ca) Impact
```

In [35]:
```
ca_group = df.groupby('ca')['condition'].mean()
```

In [36]:
```
print("Impact of the no of major vessels on persons body")
plt.figure()
plt.plot(ca_group.index, ca_group.values)
plt.title("Major Vessels vs Disease Probability")
plt.xlabel("Number of Vessels")
plt.ylabel("Disease Probability")
plt.show()
```

```
Impact of the no of major vessels on persons body
```



In [ ]:

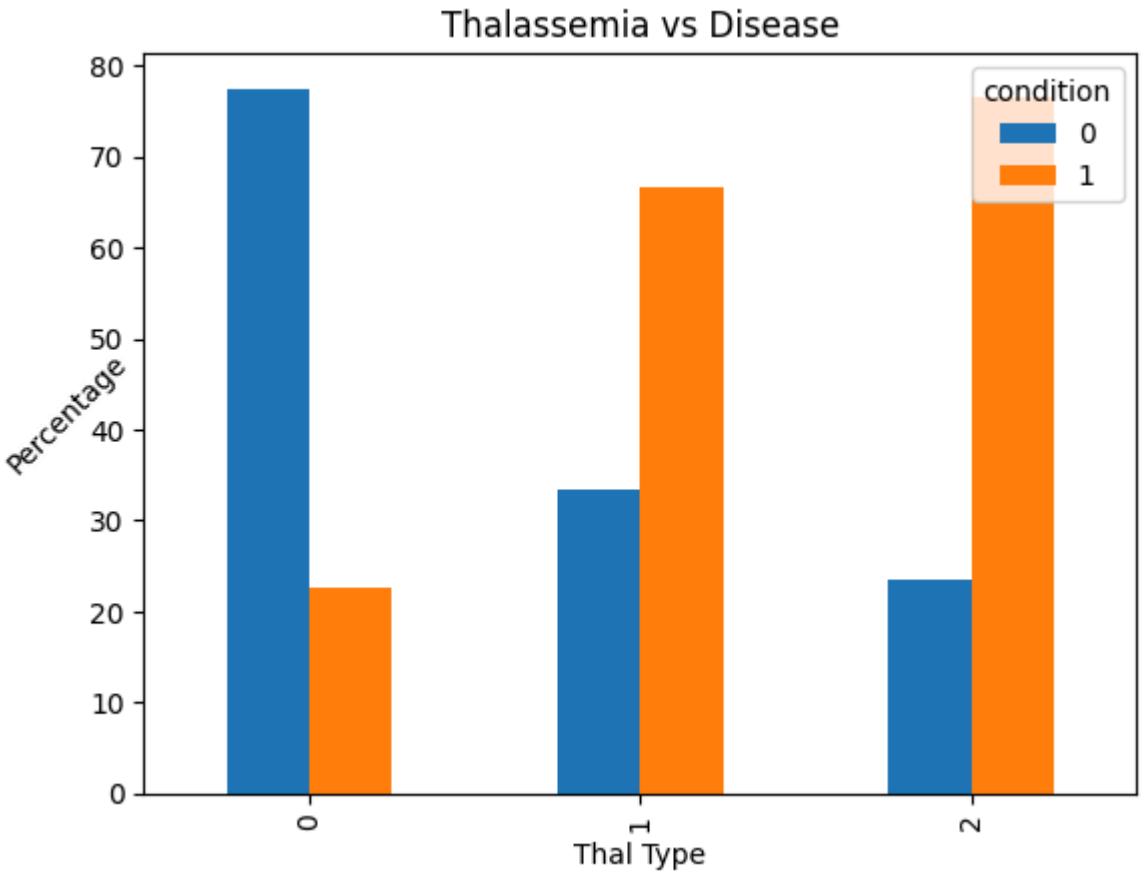
In [ ]:

In [ ]:
```
#CASE 13 : Thalassemia vs Disease
```

In [77]:
```
print("Cross-tabulate thal and target and Converting it  to percentage")
print('-----------------------------------------------')
thal_ct = pd.crosstab(df['thal'], df['condition'], normalize='index')*100
print(thal_ct)
```

```
Cross-tabulate thal and target and Converting it  to percentage
-----------------------------------------------
condition          0          1
thal
0          77.439024  22.560976
1          33.333333  66.666667
2          23.478261  76.521739
```

In [81]:
```
thal_ct.plot(kind='bar')#, stacked=True)
plt.title("Thalassemia vs Disease")
plt.xlabel("Thal Type")
plt.ylabel("Percentage")
plt.show()
```



In [ ]:

In [ ]:

In [82]:
```
#CASE 14 : Multi-Factor Risk Analysis
```

In [86]:
```
print("calculating high risk patient by using age , cholestrol, and trestbp as a p
print("                                  ")
high_risk = df[
    (df['age'] > 50) &
    (df['chol'] > 240) &
    (df['trestbps'] > 140)
]

print("Total High Risk Patients:", len(high_risk))
```

```
calculating high risk patient by using age , cholestrol, and trestbp as a parameter

Total High Risk Patients: 33
```

In [88]:
```
print("calculating the disease percent i high risk which is more dangerous")
print("Disease % in High Risk:",
      high_risk['condition'].mean()*100)
```

```
calculating the disease percent i high risk which is more dangerous
Disease % in High Risk: 66.66666666666666
```

In [ ]:

In [ ]:

In [89]:
```
#CASE 15: Create the risk score using custom analysis
```

In [95]:
```
df['risk_score'] = (df['chol']/200) + (df['trestbps']/120 + df['oldpeak'])
```

In [98]:
```
def classify(score):
    if score < 3:
        return 'LOw'
    elif score < 5:
        return "Medium"
    else:
        return "High"

df['risk_category'] = df['risk_score'].apply(classify)
risk_dist = df['risk_category'].value_counts()
```

In [53]:
```
df['risk_score'] = (df['chol'] / 200) + (df['trestbps'] / 120) + df['oldpeak']

# Classify patients as Low Risk, Medium Risk, High Risk
def classify_risk(score):
    if score < 3:
        return 'Low Risk'
    elif score <= 5:
        return 'Medium Risk'
    else:
        return 'High Risk'

df['risk_category'] = df['risk_score'].apply(classify_risk)

print("Risk score statistics:")
print(df['risk_score'].describe())
print("\nRisk category distribution:")
print(df['risk_category'].value_counts())
```

```
Risk score statistics:
count    297.000000
mean       3.389753
std        1.244547
min        1.733333
25%        2.393333
50%        3.075000
75%        4.091667
max        8.353333
Name: risk_score, dtype: float64

Risk category distribution:
risk_category
Low Risk       141
Medium Risk    122
High Risk       34
Name: count, dtype: int64
```

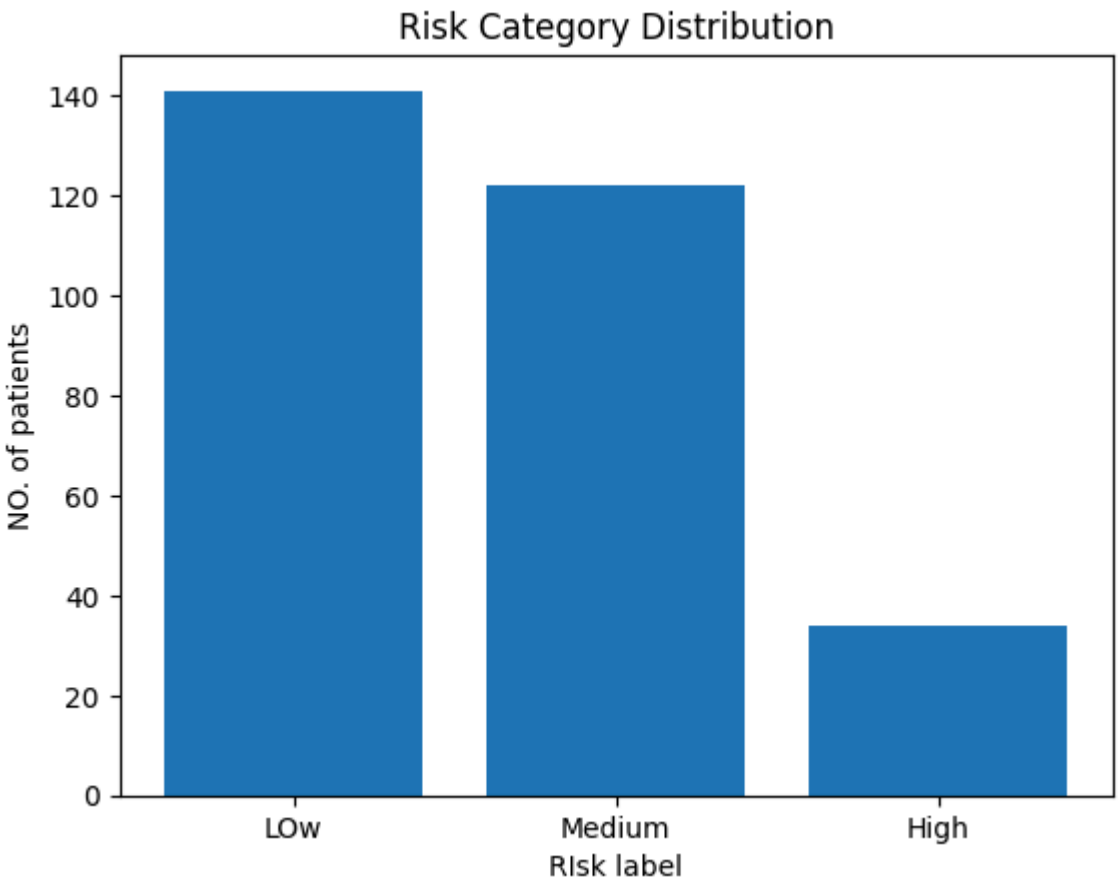
In [103]:
```
plt.figure()
```

Out[103]:
```
<Figure size 640x480 with 0 Axes>
```

```
<Figure size 640x480 with 0 Axes>
```

In [109]:
```
plt.bar(risk_dist.index, risk_dist.values)
plt.title("Risk Category Distribution")
```

```
plt.xlabel("RIsk label")
plt.ylabel("NO. of patients")
plt.show()
```



In [ ]:
```
# INSIGHTS
```

In [ ]:
```
Q1. Does cholesterol strongly impact heart disease?
->  Yes high cholesterol is a major risk factor for heart disease.
    Elevated levels of LDL cholesterol can lead to plaque buildup in arteries, nar
    Managing cholesterol through diet, exercise, and medical care is crucial for p

Q2. Is the male population more vulnerable to heart disease?
->    Men generally face a higher risk of heart disease at younger ages compared t
    While women's risk increases after menopause, men tend to develop heart proble

Q3. Does exercise-induced angina significantly increase risk?
->    Yes exercise-induced angina is a warning sign of underlying coronary artery
    It indicates that the heart is not receiving enough oxygen during physical exe
    which significantly raises the risk of serious cardiac events if left untreate

Q4. Which feature has the strongest correlation with heart disease?
->   Among the listed features, exercise-induced angina shows the strongest correl
    While age, cholesterol, and gender are important, angina directly reflects cor
    making it a powerful predictor of disease presence.
```

In [ ]: